

A Literature-Implied Partial Answer for Complex Selfadjoint L_1 -Preduals

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Status. This is a *literature-implied partial subcase*, not a claimed full solution of the complex problem in arXiv:1105.3413.

Source question. Ludvik and Spurny prove their Theorem 1.4 for real L_1 -preduals and write on page 5 that the complex case remains open because it is not clear whether Lusky's theorem that every real L_1 -predual is complemented in a simplex space remains true for complex L_1 -preduals.

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of Sections 3–6 are thus formulated for this setting. The proof of Theorem 1.4 is moreover based upon a result of W. Lusky stating that any real L_1 -predual is complemented in a simplex space (i.e., a space of type $\mathfrak{A}^c(X)$ for a simplex X) and thus our above mentioned technique can be used only for real L_1 -preduals. Since it is not clear whether Lusky's result remains true for complex L_1 -preduals, *the validity of Theorem 1.4 for complex spaces remains open.*

The content of our paper is the following. The second section provides a more detailed information on descriptive classes of sets and functions. Then we prepare a proof of Theorem 1.1 in Section 3. Results necessary for dealing with Lindelöf sets of extreme points are collected in Section 4. They are used in Sections 5 and 6, which prepares ground for the proof of Theorems 1.2 and 1.3. All Sections 3–6 deal within the context of real spaces. Section 7 proves by means of prepared results the

Implied subcase. The complex analogue of Theorem 1.4 holds for selfadjoint complex L_1 -preduals. In particular, by Hirsberg and Lazar's representation theorem, it holds for every complex Lindenstrauss space whose unit ball has an extreme point.

Theorem. *Let X be a Choquet simplex and let*

$$E = A_{\mathbb{C}}(X) := A(X) + iA(X)$$

*be the complex Banach space of continuous complex-valued affine functions on X , with the supremum norm. Let $f \in E^{**}$ be strongly affine on B_{E^*} and suppose $f \in \mathcal{C}_{\alpha}(B_{E^*})$ for some $\alpha \in [2, \omega_1)$. Then*

$$f \in \begin{cases} \mathfrak{A}_{\alpha+1}(B_{E^*}), & 2 \leq \alpha < \omega_0, \\ \mathfrak{A}_{\alpha}(B_{E^*}), & \omega_0 \leq \alpha < \omega_1. \end{cases}$$

If, in addition, the extreme boundary of the representing simplex X is Lindelöf and resolvable, then $f \in \mathfrak{A}_{\alpha}(B_{E^})$.*

Proof. Embed X into B_{E^*} by evaluations, $\delta_x(a) = a(x)$. Put $F(x) = f(\delta_x)$. Since the evaluation map is continuous, F is of Baire class α on X . Strong affinity of f on B_{E^*} implies strong affinity of F on the simplex X : if $\mu \in M^1(X)$ has barycenter x , then the push-forward of μ under $y \mapsto \delta_y$ has barycenter δ_x , hence

$$F(x) = f(\delta_x) = \int_X f(\delta_y) d\mu(y) = \int_X F(y) d\mu(y).$$

Apply Ludvik-Spurny's real simplex theorem, namely Proposition 7.1 in the proof of their Theorem 1.4, to the real and imaginary parts of F . This gives

$$F \in \begin{cases} \mathfrak{A}_{\alpha+1}(X; \mathbb{C}), & 2 \leq \alpha < \omega_0, \\ \mathfrak{A}_\alpha(X; \mathbb{C}), & \omega_0 \leq \alpha < \omega_1, \end{cases}$$

and, under the Lindelof-resolvable boundary hypothesis, $F \in \mathfrak{A}_\alpha(X; \mathbb{C})$.

It remains to lift the affine-class approximants from X back to B_{E^*} . Every $\phi \in B_{E^*}$ is represented by a complex Radon measure μ on X with $\|\mu\| \leq 1$, after extending ϕ from $E \subset C(X)$ by Hahn-Banach and applying the Riesz representation theorem. If $h \in A_{\mathbb{C}}(X)$, define $\widehat{h}(\phi) = \phi(h) = \int_X h d\mu$. If (h_n) is a bounded sequence of lower affine-class functions converging pointwise to h on X , dominated convergence gives

$$\widehat{h}_n(\phi) = \int_X h_n d\mu \longrightarrow \int_X h d\mu$$

for every $\phi \in B_{E^*}$. Thus the lifting $h \mapsto \widehat{h}$ preserves all intrinsic affine classes generated by bounded pointwise sequential limits.

Finally, the lifted function agrees with the original f . For a representing measure μ , decompose μ into a linear combination of four positive measures via the Jordan decompositions of its real and imaginary parts. After normalizing the nonzero positive parts to probability measures, strong affinity and complex linearity of $f \in E^{**}$ give

$$f(\phi) = \int_X F d\mu.$$

Hence f is exactly the lift of F , and it belongs to the asserted intrinsic affine class on B_{E^*} .

Corollary. *If a complex Lindenstrauss space E has an extreme point in its unit ball, then the complex analogue of Ludvik-Spurny's Theorem 1.4 holds for E .*

Justification. Hirsberg and Lazar proved that a complex Lindenstrauss space whose unit ball has an extreme point is isometric to $A_{\mathbb{C}}(X)$ for a Choquet simplex X . The theorem therefore applies.

Remaining open case. This does not settle the general complex L_1 -predual case. Wulbert proved that nonselfadjoint complex L_1 -preduals exist, so the missing case is a genuine residual part of the original question.

Search bounds. Cheap run indexes had no packet for arXiv:1105.3413 or this complex L_1 -predual question. Web searches for Lusky's theorem in complex L_1 -preduals and local corpus searches for "complex Lindenstrauss", "selfadjoint", and "nonselfadjoint" found the Hirsberg-Lazar and Wulbert structure results but no full later settlement of the nonselfadjoint case.

References

- [1] P. Ludvik and J. Spurny, *Descriptive properties of elements of biduals of Banach spaces*, arXiv:1105.3413, 2011.
- [2] B. Hirsberg and A. J. Lazar, *Complex Lindenstrauss spaces with extreme points*, Transactions of the American Mathematical Society 186 (1973), 141–150.
- [3] D. E. Wulbert, *Real structure in complex L_1 -preduals*, Transactions of the American Mathematical Society 235 (1978), 165–181.